

UNIT IV

Rural Housing

1] Explain about planning of rural houses?

- Rural housing in India is very imp aspect in country development.
- The main aim $\div$ provide shelter for all people in the country.
- The planning of rural housing is more complex than planning of urban housing.
- Because rural housing planning requires more administrative skills.

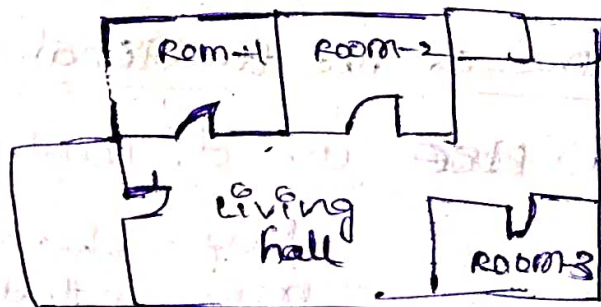
key factors (consideration)

① Building material $\div$

Materials can be selected based on their standards by considering the cost, performance, quality.

② Building requirements $\div$

- 1) plinth
- 2) ventilation
- 3) lighting
- 4) circulation
- 5) Room size
- 6) Building height.



③ Environment $\div$

- choose the suitable environment.

④ sustainable growth $\div$

If we construct a house that may be helpful in the future. [expected to see in the long term]

- i] quality
- ii] economic growth.
- iii] consistency

types of houses in rural areas :

- 1] Huts
- 2] cottages
- 3] pucca houses

⑤ consult the local authorities :

⑥ Affordability [

⑦ transportation

⑧ sustainability.

Traditional practice of rural houses :

→ Traditional practices can mainly consider the regional cultural environmental factors.

→ now-a-days these traditional practices of houses are still continues in the areas which are still under-development.

some of the traditional methods :

① use of local materials :

clay, bamboo

mud, thatch,

grass, stone,

wood

- less cost

well-suited to climate conditions

② mud & clay construction

adobe, cob, rammed earth.

③ thatched roof construction

thatch, palm leaves

grass

straw

④ wood house construction [timber]

⑤ stone-house construction.

⑥ bamboo house construction.

Importance ÷

- 1) cultural preservation.
- 2) cost-effective.
- 3) community cohesion [ownership, co-operation, support].
- 4) climate adaptation.
- 5) environmental sustainability.

advantages ÷

- 1] less-cost
- 2] local-sourcing materials
- 3] No transportation costs
- 4] cultural preservation.
- 5] climate adaptation. [adaptability]
- 6] sustainability.
- 7] Availability
- 8] Accessibility.
- 9] Environmental Eco-friendly

dis-advantages ÷

Limited.

→ durability

→ maintenance

→ Aesthetics. [

→ Technological limitations.

→ lack of skill

restore earth-quake damaged buildings

- ① seismic strengthening
- ② structural repairing ^{best repairing materials, techniques} [cracked walls,
- ③ Retrofitting ^{modifying} [strength & stiffness] ^{performance}
- ④ grouting [cement, epoxy].
- ⑤ providing extra reinforcement [steel-iron reinforcement]
- ⑥ choose the safest place for construction.

① seismic strengthening :-

→ It is the technical intervention that improves seismic resistance by

improving the

* strength

* stiffness

* ductility of the building.

② structural repairing :-

* best structural materials

- 1) cracked walls
- 2) Ferrocement
- 3) epoxy mortar
- 4) polymer concrete
- 5) shear wall
- 6) steel bracing.
- 7) sticking walls.

mud-housing techniques = mud roofing.

→ ancient housing techniques

→ environment eco-friendly.

→ mainly rural-type housing technique.

1) Adobe → Adobe bricks → clay, mud, sand, water, straw, some stabilizers.

2) Cob → mixture of sand, water, clay, & straw (curved structure).

3) Rammed earth → [ancient tech] mixture of gravel, sand, silt, clay.

4) Wattle & Daub → available in blocks.

→ wooden strips (walls).

→ thousands of years

Daub = sticky material.

→ made up of animal dung, clay, soil, sand, straw.

→ weather proof.

→ layer-by-layer.

→ houses & museums.

→ Each layer is maximum of 15cm.

Advantages :-

- less cost
- sustainability
- environmental impact
- Affordability
- insulation properties

mud-based roofs :-

disadvantages

- durability
- maintenance
- Aesthetics
- weather vulnerability.

← simplicity
sustainability
insulation properties.

Thatched roofs → thatch, straw, palm leaves, grass, bamboo, wood.

Mud-based roofs → some times mud can be shaped into tiles.

characteristics of mud :-

- 1] composition
- 2] sustainability
- 3] local availability
- 4] adaptability
- 5] durability
- 6] Thermal properties
- 7] insulation properties
- 8] consistency

uses :-

- construction
- pottery [vessels, pots]
- ceramics
- Agriculture.

Fire treatment for thatch roof :-

- thatched roof :- thatch, straw, palm leaves, bamboo, wood, dry grass
- safety measure } improves the life-time of house

Some of the fire retardant treatments

1) Natural fire retardants :-

- hydrated lime
- clay mixtures

2) chemical treatments

- chemical sprays
- makes the thatch inflammable
-

3) Borate treatment :-

- insect repellent [sod wood]
- applied for wood
- Avoid the rusting.
- It is reapplied after 6-10 years.
- used to control ants, cockroaches & insecticides.

4) Thatch batts.

5) Aluminium barrier foil

6) Thatch fire-retardant blanket.

considerations :-

- 1] compatability -
- 2] Reapplication -
- 3] durability -
- 4] Regulations -
- 5] ^{safety} standards -
- 6] Effectiveness -
- 7] Professional guidance.

Advantages :-

- Improves consistency
- " durability.
- safety enhancement.
- Improve fire-resistance
-

disadvantages :-

- cost effective
- environmental impact
- sustainability
- maintenance requirement.

Soil-stabilization :-

- increasing the properties of soil by eliminating

Pathogens
Pests
Contaminants
micro-organisms

}

these compounds from the soil.
- Agriculture
horticulture
Construction houses

}

*) quality of soil.
*) strength of soil.

Methods :-

- 1] heat treatment
 - 2] chemical treatment
 - 3] solarisation [plastic] / using solar energy.
 - 4] crop rotation
 - 5] Biological ctrl [microorganisms]
[pesticides]
 - 6] composting.
 - 7] cement stabilization,
 - 8] lime stabilization
 - 9] fly-ash stabilization
- ## Importance of soil stabilization

- 1] Improve soil strength
- 2] survive from soil erosion.
- 3] Improve durability
- 4] Cost effectiveness.
- 5] Environmental benefits.

Advantages :-

- Improve bearing capacity
- Avoid swelling & shrinkages
- plant growth
- economic growth.
- saves cost.
- strength stability.
- soil erosion (save from)
- environmental ecofriendly.

dis-advantages

- 1] waste of soil
- 2] expensive
- 3] required soil testing
- 4] contaminants wastes.

Rural housing programs :-

- ① Pradhan mantri Awas Yojana [2016];
- ② Indira Awas Yojana [1996]
scheduled castes (SC) /
scheduled tribes (ST) /
→ Below poverty line (BPL).
- ③ RLEGP (Rural Landless employment Guarantee program).
- ④ Rajiv Awas Yojana (slum areas)

Importance :- & Advantages :-

- Avoid poverty
- Houses for All.
- Rural development. [culture, society, economy]
- Infrastructure development
- health [improper houses lead]
- Reduce homelessness.

dis-advantages :-

- 1) lack of co-ordination
- 2] Inequality
- 3] low investment
- 4] Infrastructure. [
- 5] Demography [statistical study of human population]